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# WEB SISTEMAK

## 2025/2026 IKASTURTEA

Web zerbitzariak  
Garapen software-aren instalazioa



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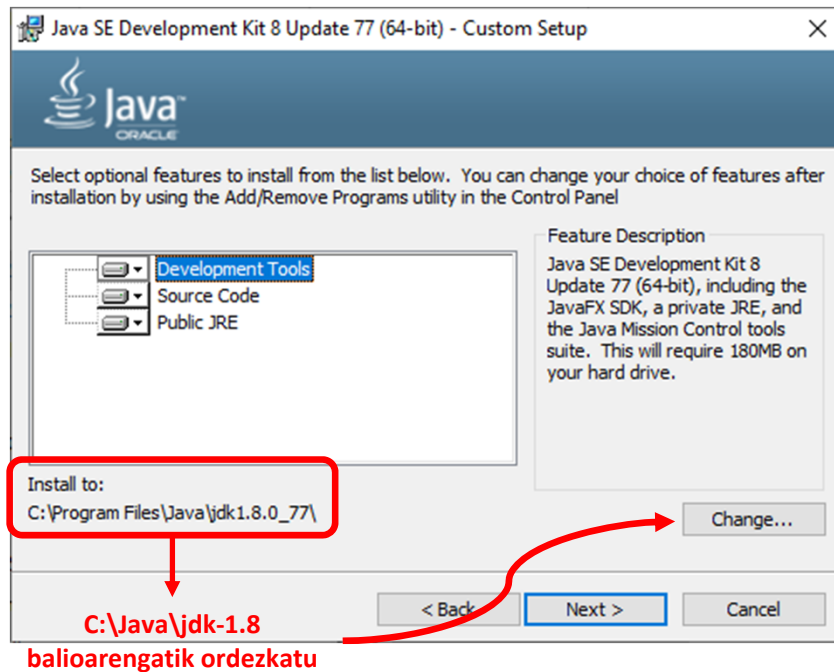
# INSTALATU BEHARREKO SOFTWARE-A

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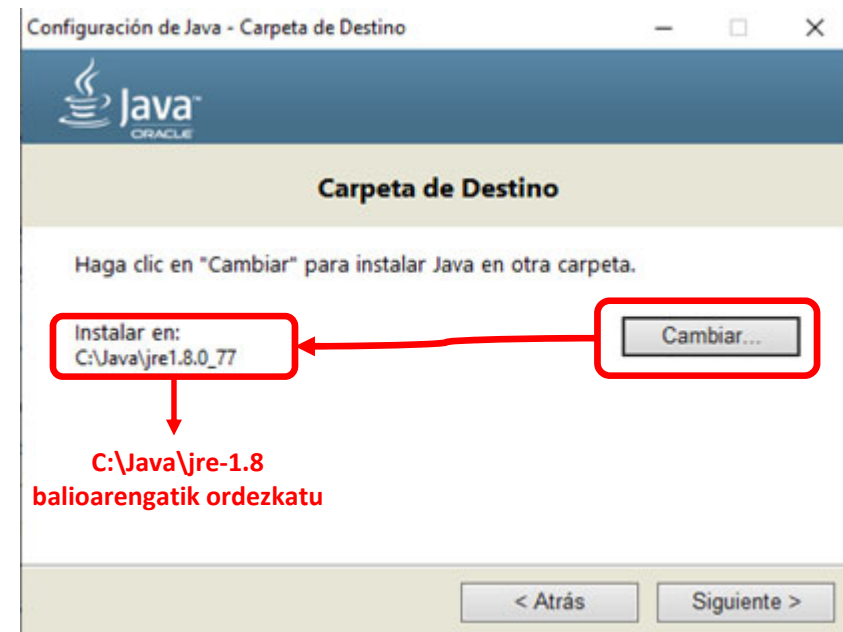
- Java Development Environment (JDK)
  - JDK **C:\Java\jdk-1.8** direktorioan instalatu ezazue
  - JRE **C:\Java\jre-1.8** direktorioan instalatu ezazue
  - **JAVA\_HOME** ingurune aldagaia sortu, bere balioa **C:\Java\jdk-1.8** delarik
- IntelliJ
  - Instalatzaileak proposatutako direktorioan instalatu ezazue
- MySQL datu-base kudeatzailea
  - Setup type: *Custom*.
  - MySQL server configuration
    - Erabiltzailea: *root*. Pasahitza: *root*
    - Windows service (datu-basea zerbitzu bezela abiarazi daitela Windows-en abiaraztearekin batera)
      - *Configure MySQL server as a Windows service*
      - *Start MySQL server at system startup*
- Apache Tomcat web zerbitzaria
  - **C:\apache-tomcat-8.5.11** direktorioan instalatu ezazue
  - **CATALINA\_HOME** ingurune aldagaia sortu, bere balioa **C:\apache-tomcat-8.5.11** delarik
  - Egiaztapena:
    - cmd → C:\apache\_tomcat\bin\startup.bat
    - Nabigatzailean → <http://localhost:8080/>

Windows erabiltzen duzuenontzako,  
programak ehubox-en bilduta daude:  
<https://ehubox.ehu.eus/s/x98mFPJbSi2oJ5N>

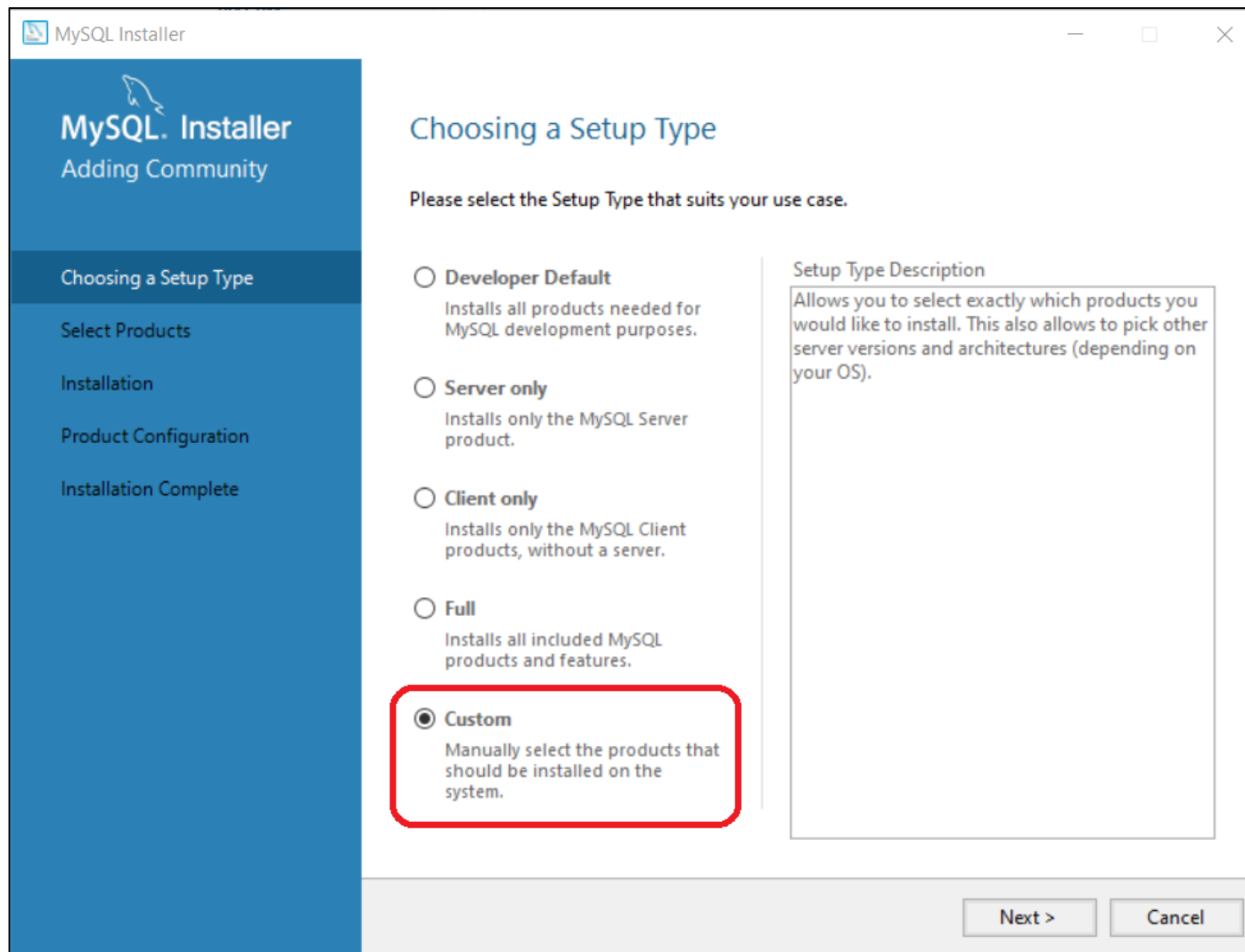
# JDK



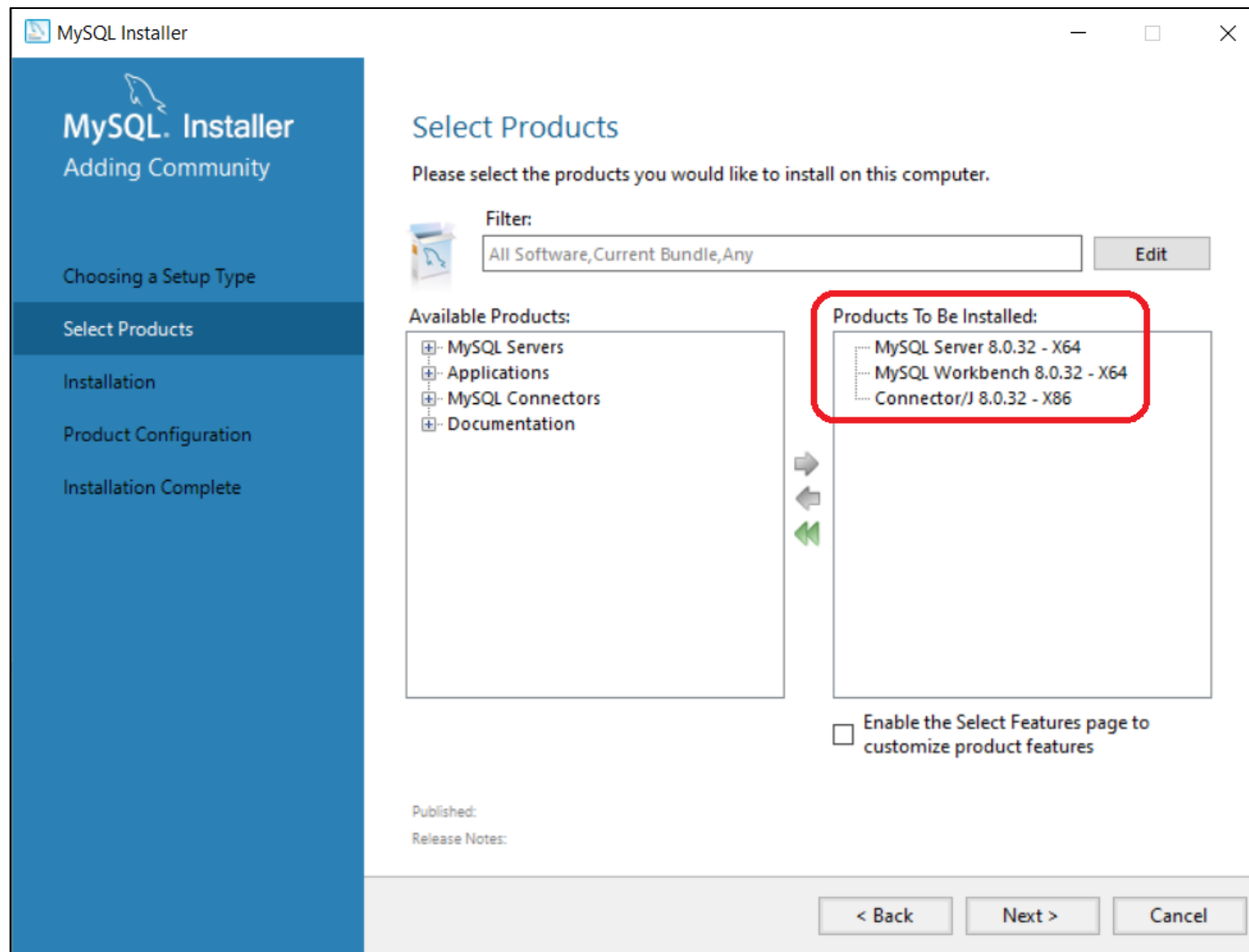
**JRE C:\Java\jre-1.8**  
direktorioan instalatu



# MYSQL INSTALLER



# MySQL INSTALLER



# MySQL INSTALLER

MySQL Installer  
MySQL Server 8.0.32

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Server File Permissions

Apply Configuration

### Type and Networking

#### Server Configuration Type

Choose the correct server configuration type for this MySQL Server installation. This setting will define how much system resources are assigned to the MySQL Server instance.

Config Type: Development Computer

#### Connectivity

Use the following controls to select how you would like to connect to this server.

☒ TCP/IP Port: 3306 X Protocol Port: 33060

☒ Open Windows Firewall ports for network access

☐ Named Pipe Pipe Name: MYSQL

☐ Shared Memory Memory Name: MYSQL

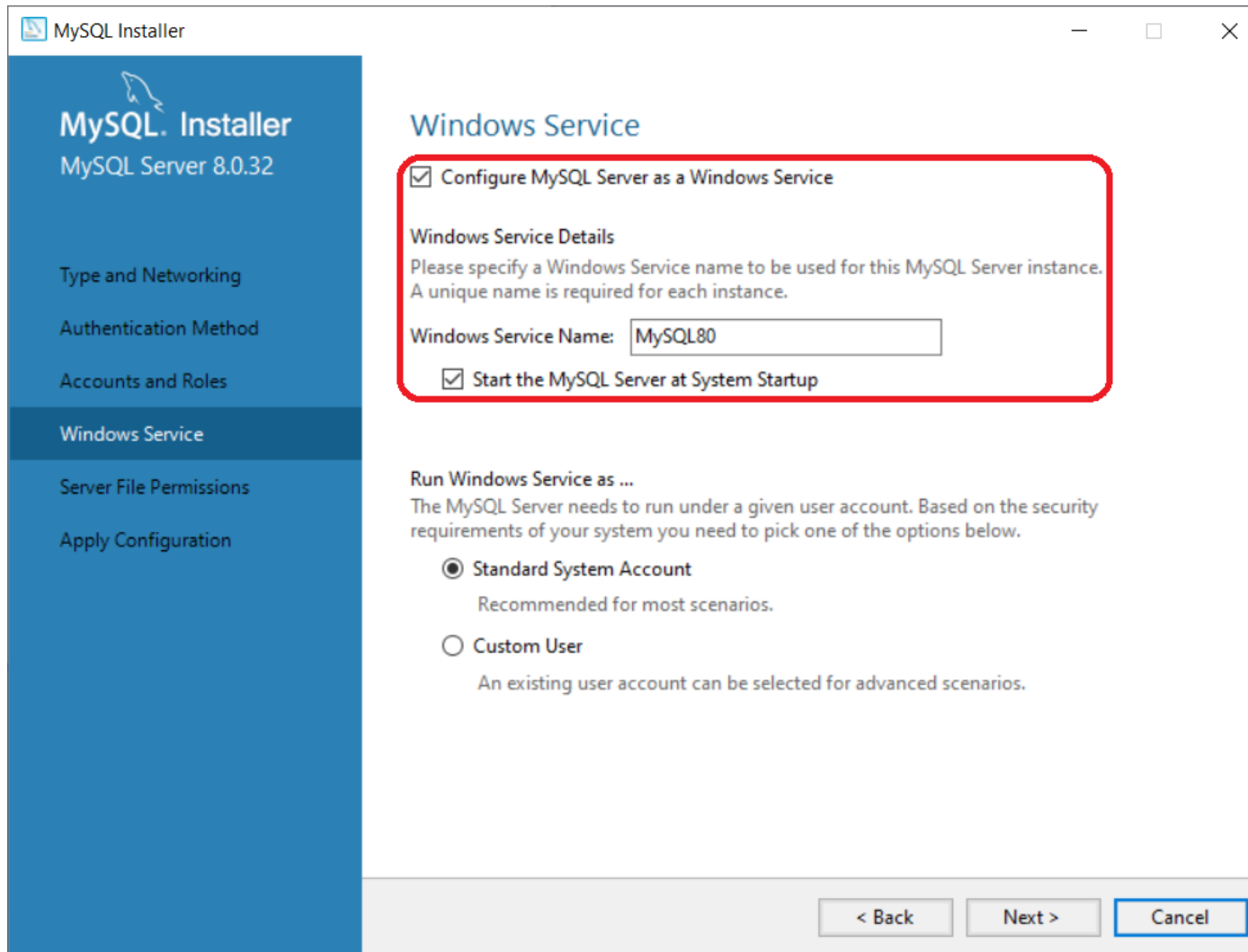
#### Advanced Configuration

Select the check box below to get additional configuration pages where you can set advanced and logging options for this server instance.

☐ Show Advanced and Logging Options

Next > Cancel

# MySQL INSTALLER



The screenshot shows the 'MySQL Installer' window for 'MySQL Server 8.0.32'. The left sidebar contains the following steps: 'Type and Networking', 'Authentication Method', 'Accounts and Roles', 'Windows Service' (which is the active step), 'Server File Permissions', and 'Apply Configuration'. The main area is titled 'Windows Service' and contains the following options:

- ☒ **Configure MySQL Server as a Windows Service**

**Windows Service Details**  
Please specify a Windows Service name to be used for this MySQL Server instance. A unique name is required for each instance.

Windows Service Name:

- ☒ **Start the MySQL Server at System Startup**

**Run Windows Service as ...**  
The MySQL Server needs to run under a given user account. Based on the security requirements of your system you need to pick one of the options below.

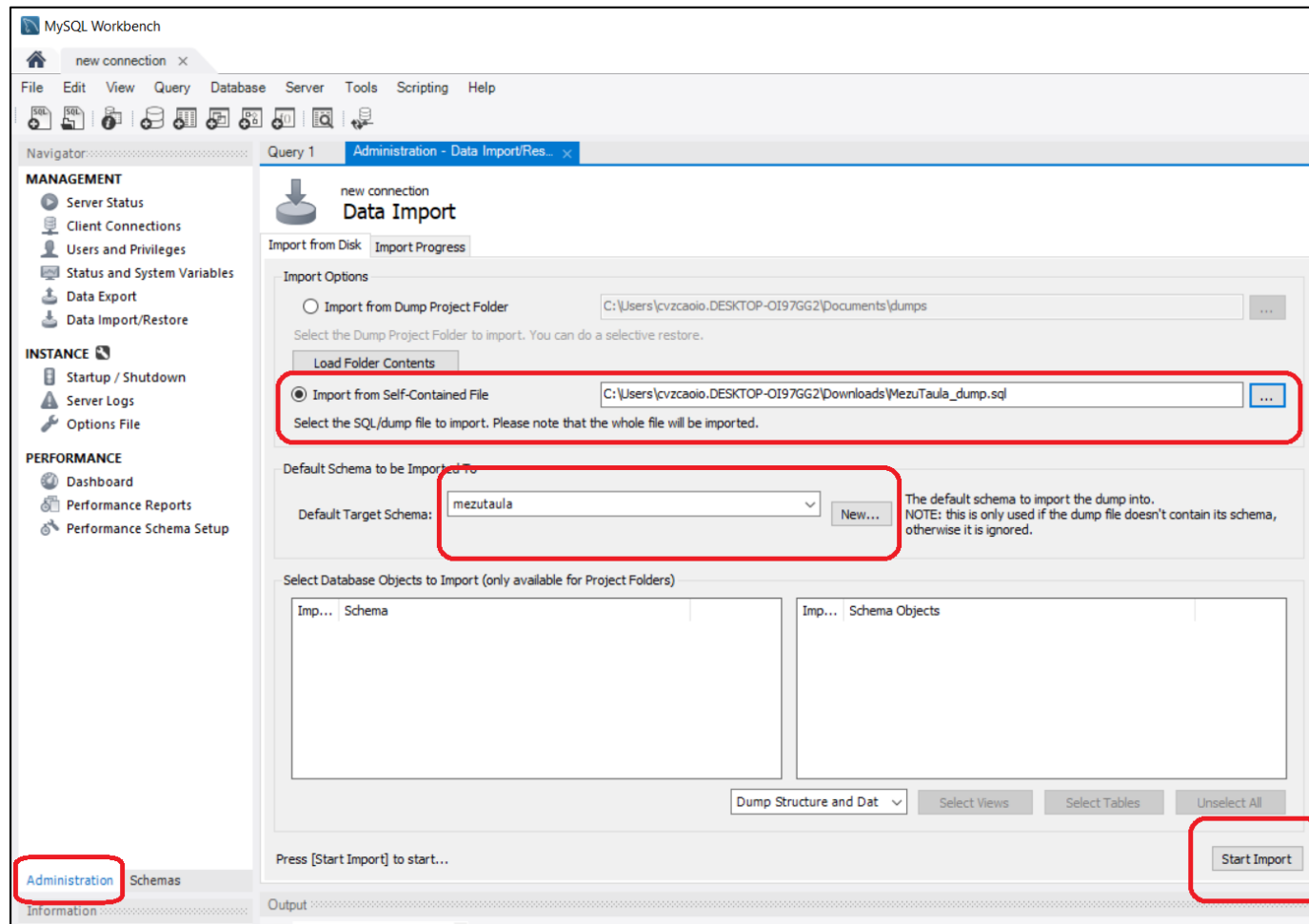
- ☒ **Standard System Account**  
Recommended for most scenarios.
- ☐ **Custom User**  
An existing user account can be selected for advanced scenarios.

At the bottom right, there are three buttons: '< Back', 'Next >', and 'Cancel'.

# MySQL ZERBITZARIAREN EGIAZTAPENA:

## DATUBASE BAT INPORTATU

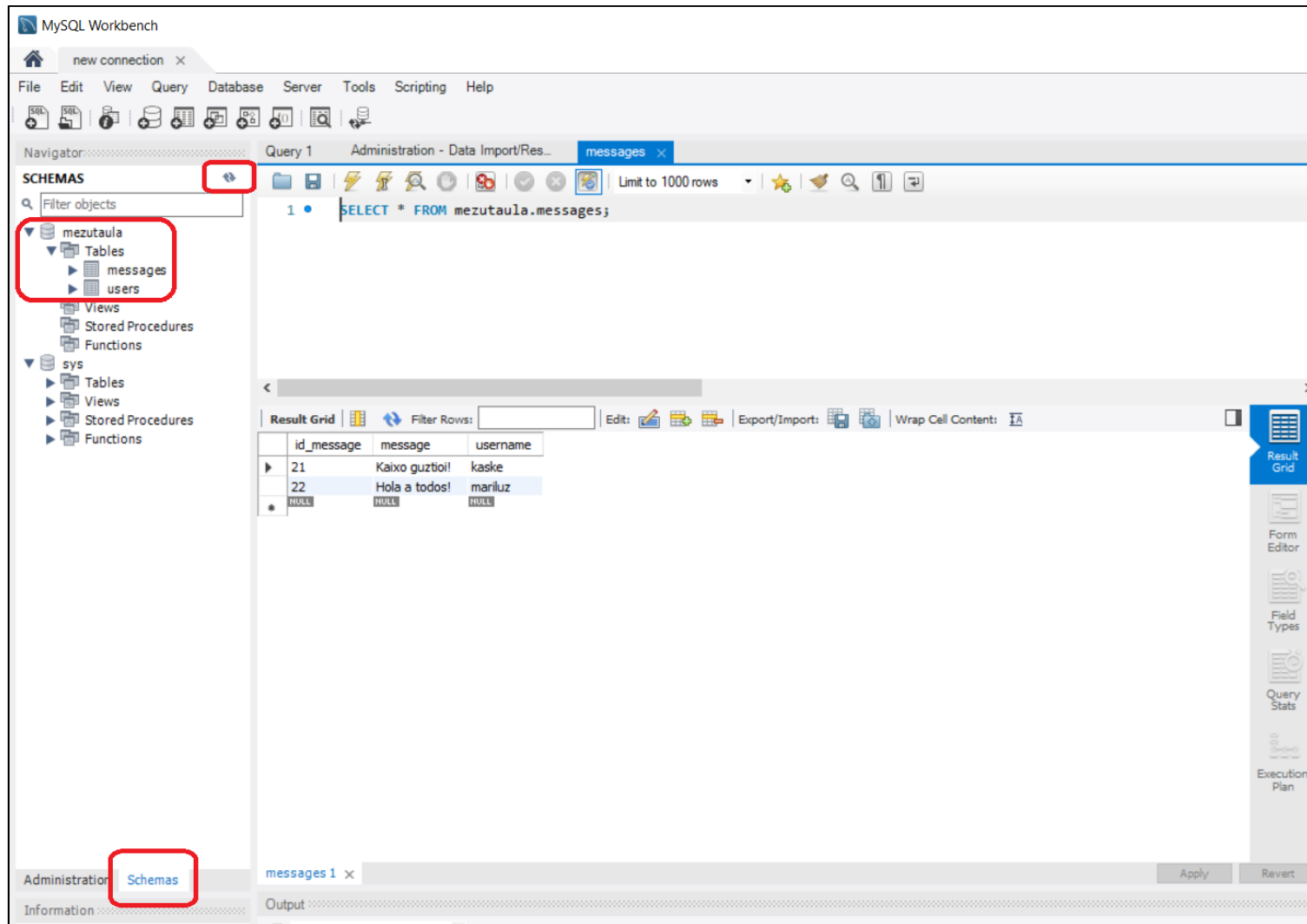
- Datu basearen iraulketa fitxategia eGelan dago.





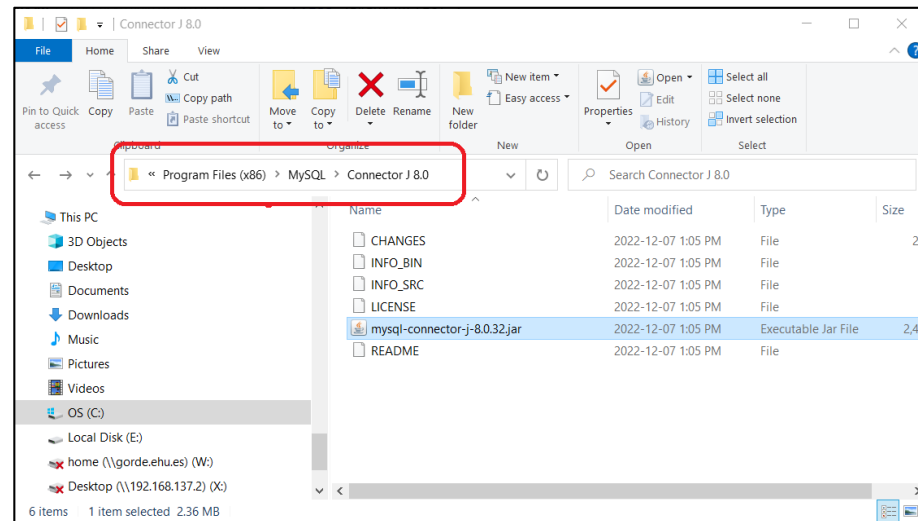
# MYSQL ZERBITZARIAREN EGIAZTAPENA:

## DATUBASE BAT INPORTATU



# JAVA ETA MYSQL ARTEKO KONEXIOA EGIAZTATU

- MySQL konektorea C:\Java\lib direktorioan kopiatu:

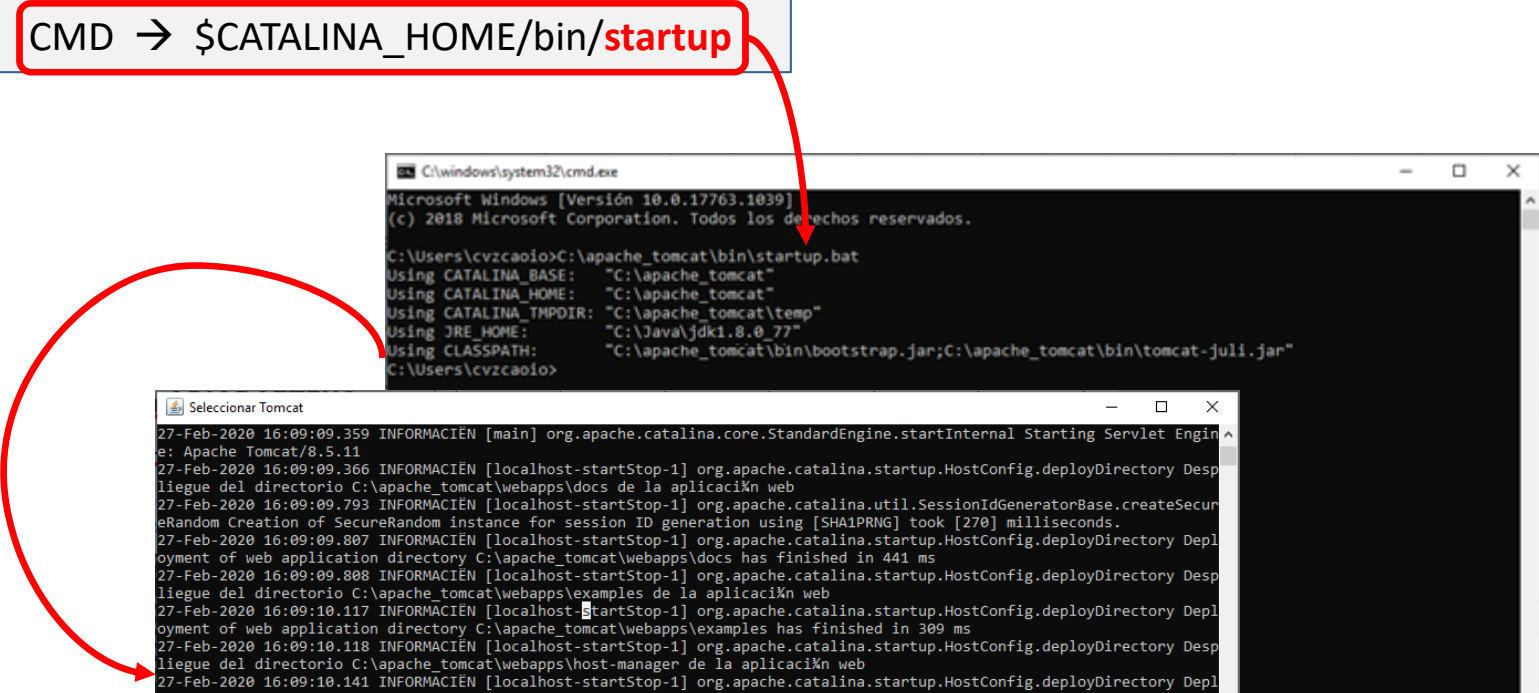


- ARIKETA:** IntelliJ-en datu basera konektatzen den Java main() bat programatu.
  - Java motako proiektu berria sortu
  - src azpidirektorioan *JavaMySQLtest.java* kopiatu (programa eGelan dago).
  - Konpilatu. Konpilazioak errorerik eman al du?
  - Exekutatu. Exekuzioak errorerik eman al du? Zergaitik?  
Zer egin behar da arazoa konpontzeko?

# TOMCAT: ABIARAZTEA

## Tomcat web zerbitzaria ABIARAZTEKO:

CMD → \$CATALINA\_HOME/bin/startup



```
C:\windows\system32\cmd.exe
Microsoft Windows [Versión 10.0.17763.1039]
(c) 2018 Microsoft Corporation. Todos los derechos reservados.

C:\Users\cvzcaoi>C:\apache_tomcat\bin\startup.bat
Using CATALINA_BASE: "C:\apache_tomcat"
Using CATALINA_HOME: "C:\apache_tomcat"
Using CATALINA_TMPDIR: "C:\apache_tomcat\temp"
Using JRE_HOME: "C:\Java\jdk1.8.0_77"
Using CLASSPATH: "C:\apache_tomcat\bin\bootstrap.jar;C:\apache_tomcat\bin\tomcat-juli.jar"
C:\Users\cvzcaoi>
```

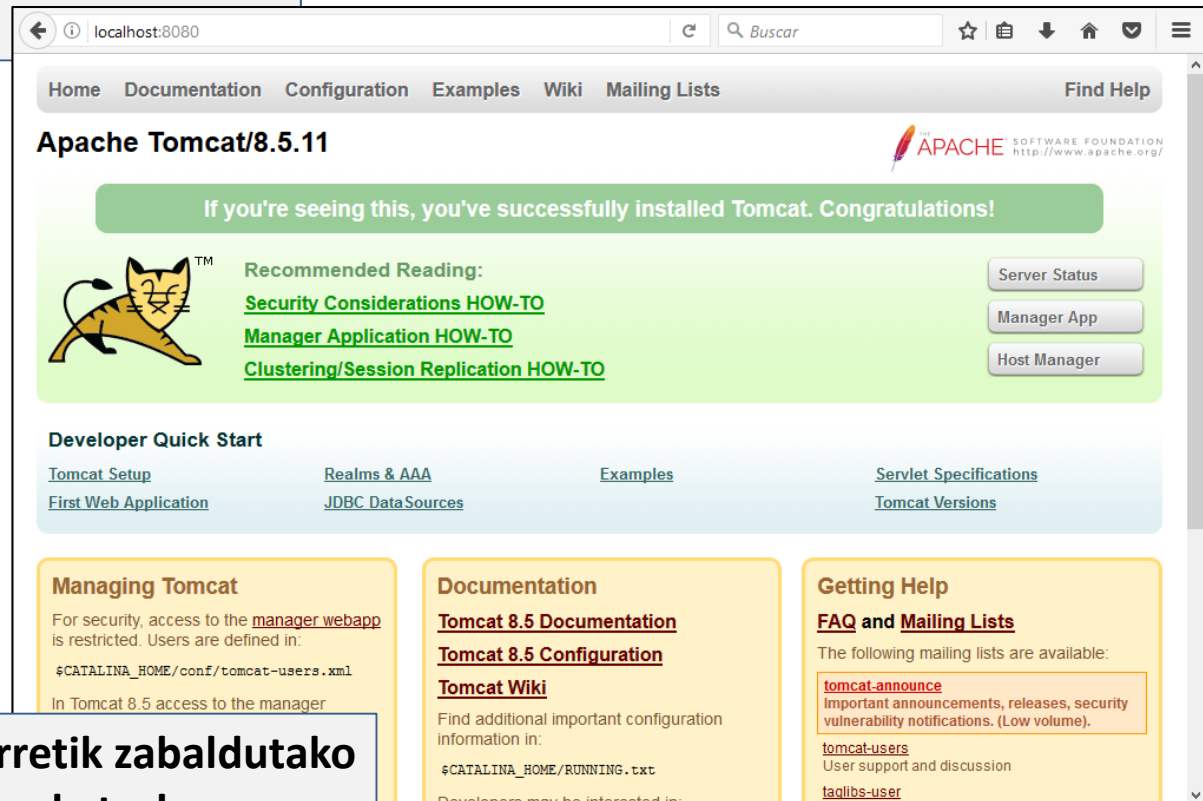
```
Seleccionar Tomcat
27-Feb-2020 16:09:09.359 INFORMACIÉN [main] org.apache.catalina.core.StandardEngine.startInternal Starting Servlet Engine
e: Apache Tomcat/8.5.11
27-Feb-2020 16:09:09.366 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Desp
liegue del directorio C:\apache_tomcat\webapps\docs de la aplicaciñ web
27-Feb-2020 16:09:09.793 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.util.SessionIdGeneratorBase.createSecur
eRandom Creation of SecureRandom instance for session ID generation using [SHA1PRNG] took [270] milliseconds.
27-Feb-2020 16:09:09.807 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Depl
oyment of web application directory C:\apache_tomcat\webapps\docs has finished in 441 ms
27-Feb-2020 16:09:09.808 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Desp
liegue del directorio C:\apache_tomcat\webapps\examples de la aplicaciñ web
27-Feb-2020 16:09:10.117 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Depl
oyment of web application directory C:\apache_tomcat\webapps\examples has finished in 309 ms
27-Feb-2020 16:09:10.118 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Desp
liegue del directorio C:\apache_tomcat\webapps\host-manager de la aplicaciñ web
27-Feb-2020 16:09:10.141 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Depl
oyment of web application directory C:\apache_tomcat\webapps\host-manager has finished in 22 ms
27-Feb-2020 16:09:10.141 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Desp
liegue del directorio C:\apache_tomcat\webapps\manager de la aplicaciñ web
27-Feb-2020 16:09:10.162 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Depl
oyment of web application directory C:\apache_tomcat\webapps\manager has finished in 21 ms
27-Feb-2020 16:09:10.163 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Desp
liegue del directorio C:\apache_tomcat\webapps\ROOT de la aplicaciñ web
27-Feb-2020 16:09:10.180 INFORMACIÉN [localhost-startStop-1] org.apache.catalina.startup.HostConfig.deployDirectory Depl
oyment of web application directory C:\apache_tomcat\webapps\ROOT has finished in 17 ms
27-Feb-2020 16:09:10.182 INFORMACIÉN [main] org.apache.coyote.AbstractProtocol.start Starting ProtocolHandler [http-nio-
8080]
27-Feb-2020 16:09:10.187 INFORMACIÉN [main] org.apache.coyote.AbstractProtocol.start Starting ProtocolHandler [ajp-nio-8
009]
27-Feb-2020 16:09:10.188 INFORMACIÉN [main] org.apache.catalina.startup.Catalina.start Server startup in 846 ms
```

Zerbitzariaren abiaraztearen arazketarako irteera mezuak (ez da errorerik agertu behar)

# TOMCAT: EGIAZTAPENA

## Tomcat ondo abiarazi dela egiaztatzeko:

Nabigatzailean <http://localhost:8080> URI-a eskatu eta Tomcat-en ongietorri orria bistaratzen dela egiaztatu.



Egiaztapena egin ostean, aurretik zabaldutako terminalak itxi 8080. portua askatzeko.

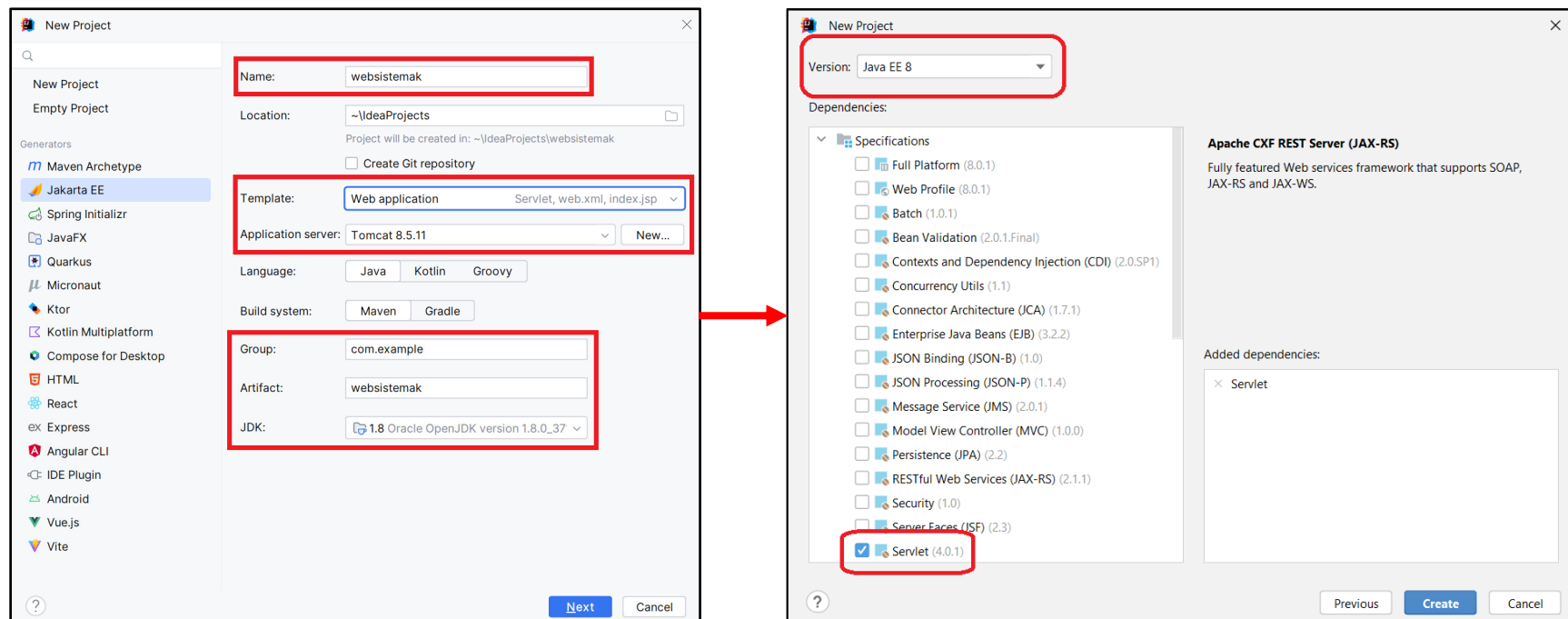
# TOMCAT: INTELIJ-EKIN INTEGRATU

Tomcat IntelliJ-tik zuzenean erabiltzeko:

Integration with application servers:

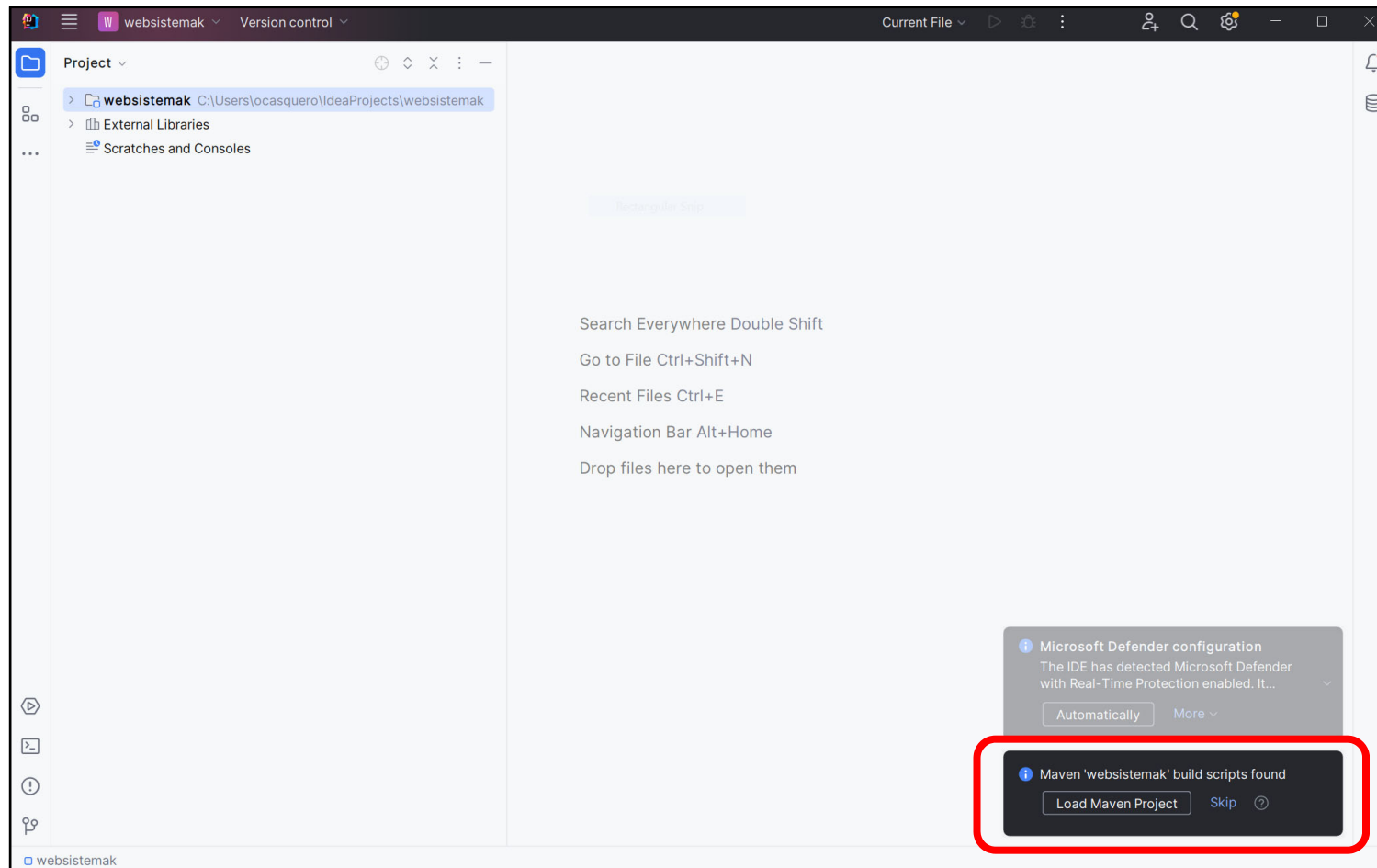
<https://www.jetbrains.com/help/idea/configuring-and-managing-application-server-integration.html>

IntelliJ-en web aplikazioak sortzeko:

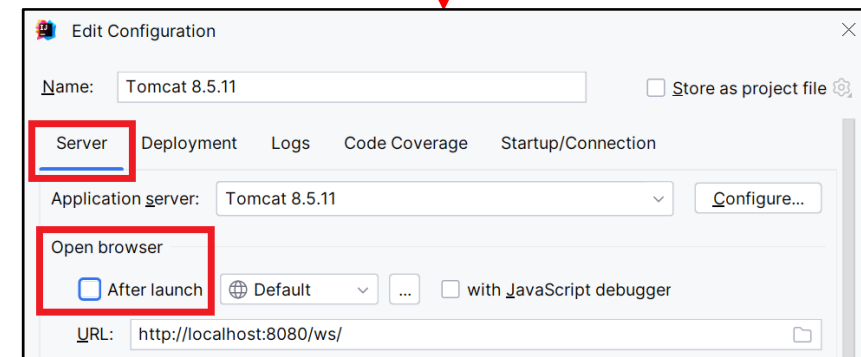
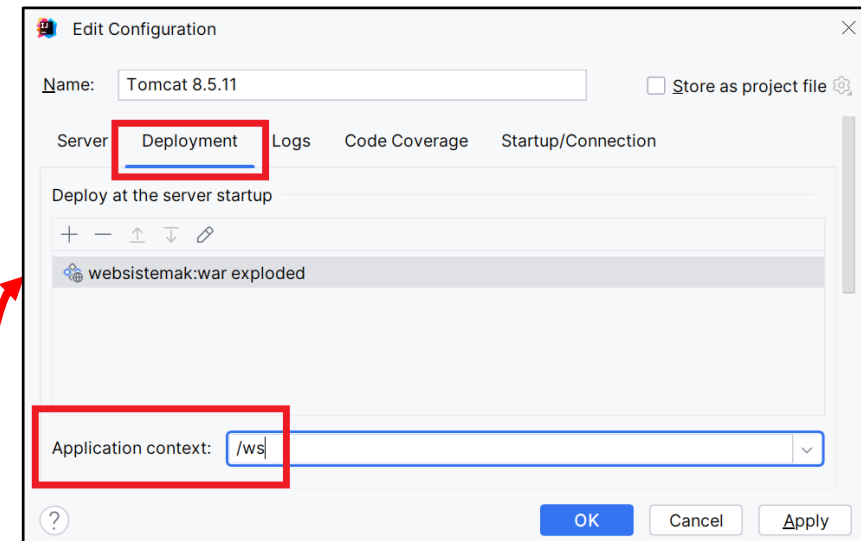
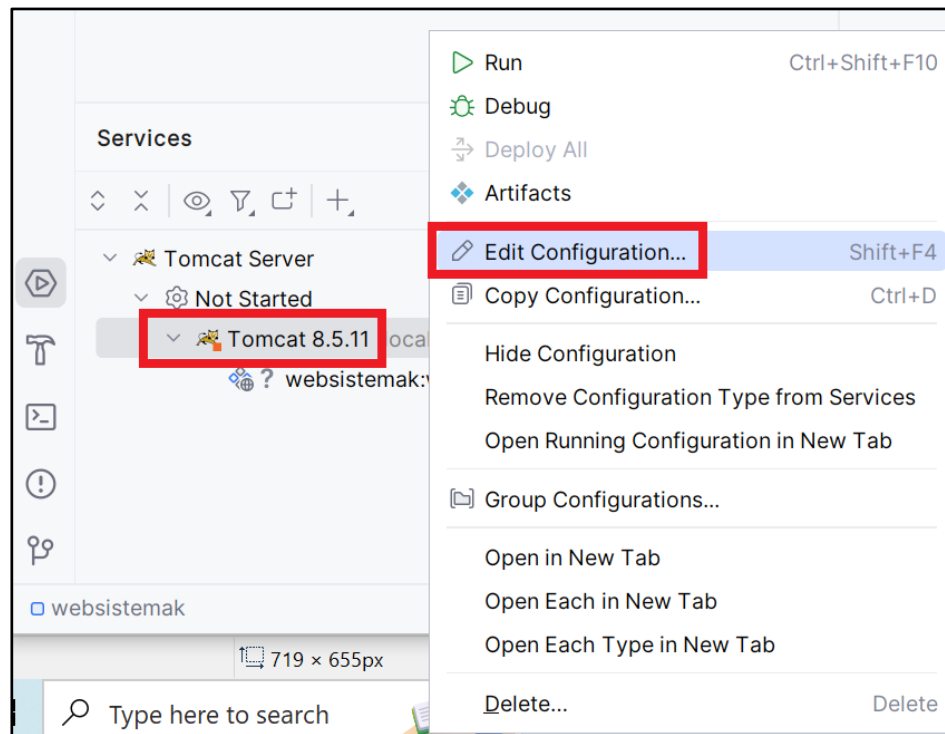


# TOMCAT: INTELIJ-EKIN INTEGRATU

Maven script eraikitzaileak kargatu Tomcat proiektu motarentzako:

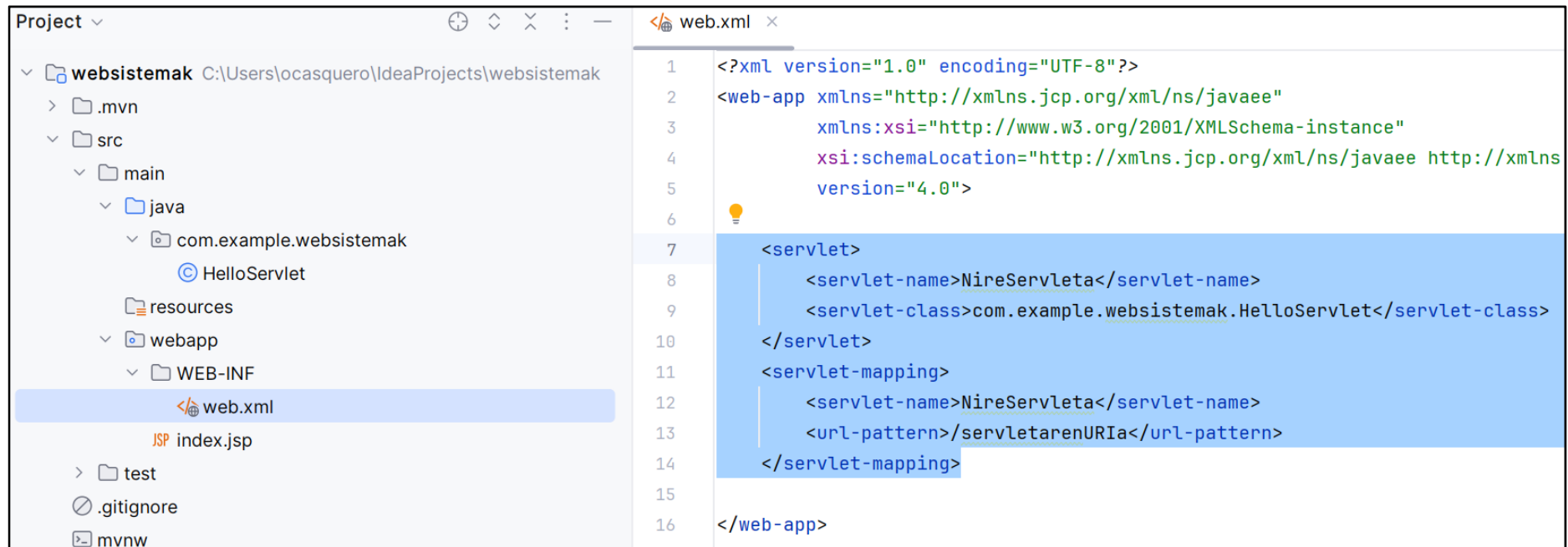


# TOMCAT: PROIEKTU BATEN URIAREN EGOKITZAPENA



# TOMCAT: PROBA

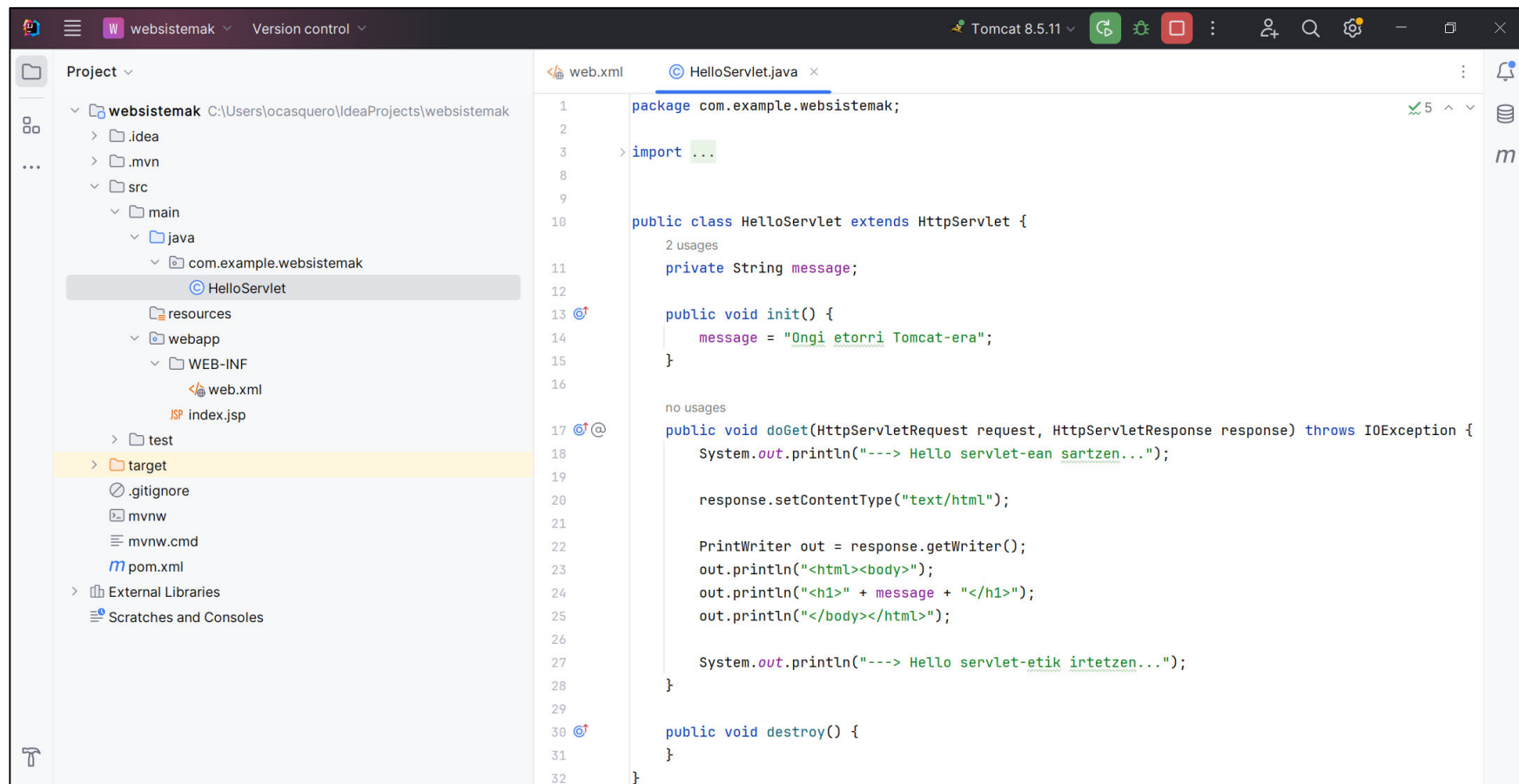
- Sortutako websistemak proiektuan:
  - web.xml fitxategian ondorengo edukia gehitu (edukia eGelatik hartu):





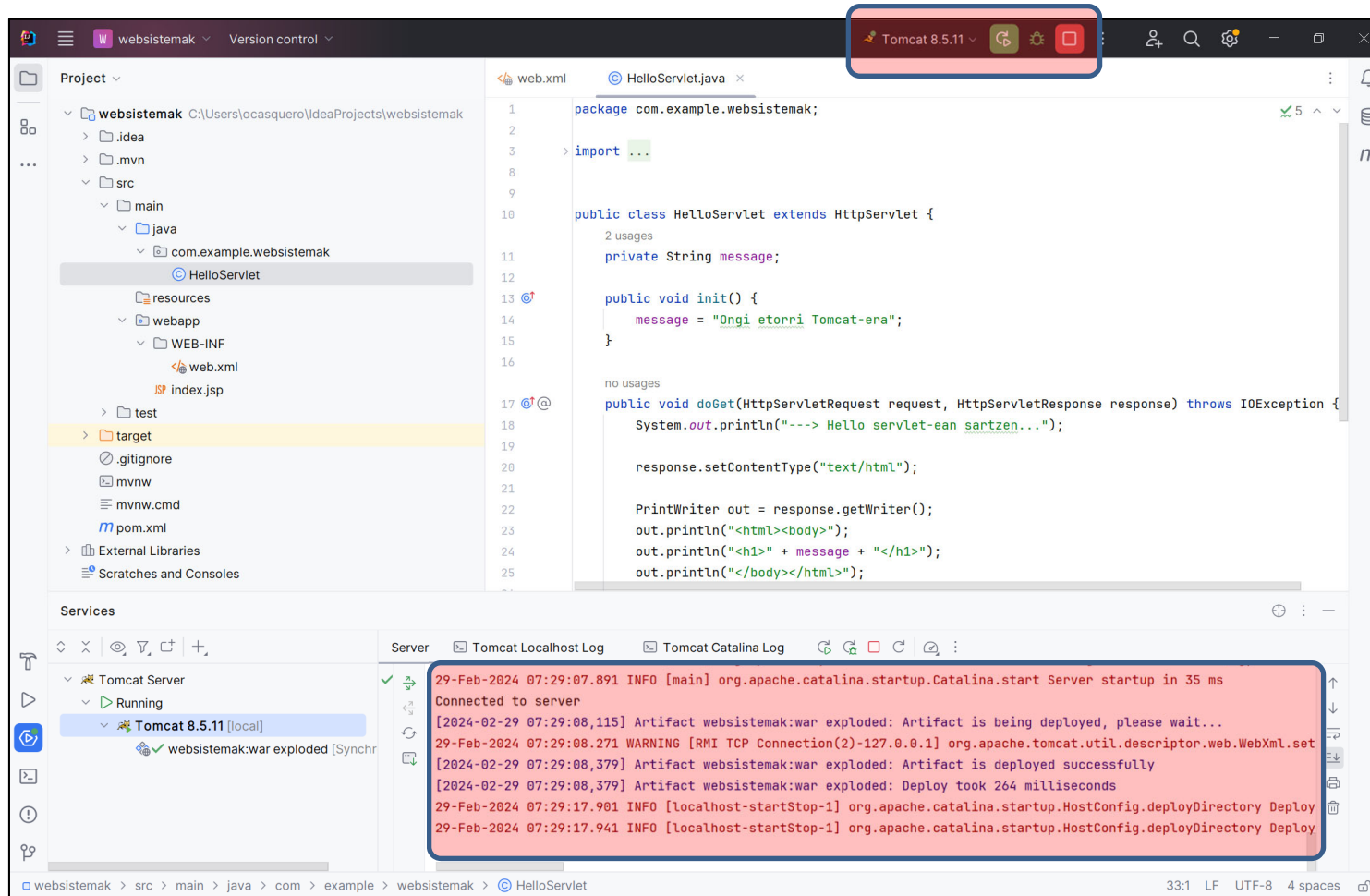
# TOMCAT: PROBA

- Sortutako websistemak proiektuan:
  - Hello servlet-aren edukia eGela-n dagoenarengatik ordezkatu:



# TOMCAT: PROBA

- IntelliJ-etik Tomcat abiarazi:



# TOMCAT: PROBA

- Nabigatzailean <http://localhost:8080/ws/servletarenURLa> zabaldu:

The screenshot shows an IDE (IntelliJ IDEA) with the `HelloServlet.java` file open. The code is as follows:

```
10 public class HelloServlet extends HttpServlet {
11     2 usages
12     private String message;
13
14     public void init() {
15         message = "Ongi etorri Tomcat-era";
16     }
17
18     no usages
19     public void doGet(HttpServletRequest request, HttpServletResponse response) throws IOException {
20         System.out.println("---> Hello servlet-ean sartzen...");
21
22         response.setContentType("text/html");
23
24         PrintWriter out = response.getWriter();
25         out.println("<html><body>");
26         out.println("<h1> " + message + "</h1>");
27         out.println("</body></html>");
28
29         System.out.println("---> Hello servlet-etik irtetzen...");
30     }
31 }
```

Red arrows indicate the flow of data from the code to the browser. The first arrow points from the `message` assignment in the `init()` method to the `message` variable in the `doGet()` method. The second arrow points from the `doGet()` method to the browser window. The browser window shows the URL `localhost:8080/ws/servletarenURLa` and the output `Ongi etorri Tomcat-era`.

The IDE's Services tab shows the Tomcat 8.5.11 server running. The Tomcat Catalina Log shows the following messages:

```
[2024-02-29 07:32:29,592] Artifact websistemak:war exploded: Artifact is
29-Feb-2024 07:32:29,703 WARNING [RMI TCP Connection(2)-127.0.0.1] org.ap
[2024-02-29 07:32:29,824] Artifact websistemak:war exploded: Artifact is
[2024-02-29 07:32:29,835] Artifact websistemak:war exploded: Deploy took
29-Feb-2024 07:32:39,213 INFO [localhost-startStop-1] org.apache.catalina
29-Feb-2024 07:32:39,212 INFO [localhost-startStop-1] org.apache.catalina
---> Hello servlet-ean sartzen...
---> Hello servlet-etik irtetzen...
```